

CSS Word-Wrap & Word-Break

Complete Guide: Controlling Text Wrapping and Breaking

Contents

1	Introduction	2
1.1	The Word Breaking Problem	2
1.2	When Word Breaking Matters	2
2	overflow-wrap Property	3
2.1	overflow-wrap Values	3
2.2	overflow-wrap Examples	3
2.3	word-wrap (Deprecated)	3
3	word-break Property	4
3.1	word-break Values	4
3.2	word-break Examples	4
4	Comparing word-break and overflow-wrap	4
4.1	Behavior Differences	4
4.2	Detailed Examples	4
5	Real-World Use Cases	6
5.1	Use Case 1: Chat Application	6
5.2	Use Case 2: Code Block Display	6
5.3	Use Case 3: Email Address Display	6
5.4	Use Case 4: Username Display	7
5.5	Use Case 5: Responsive Container	7
6	Combining with white-space	8
6.1	white-space Interactions	8
6.2	Combining Examples	8
7	Hyphens and word-break	9
7.1	Hyphens Integration	9
8	CJK (Chinese, Japanese, Korean) Text	9
8.1	CJK-Specific Properties	9

9	Browser Support	11
9.1	overflow-wrap and word-break Support	11
9.2	Fallback Strategies	11
10	Performance Considerations	11
10.1	Performance Impact	11
10.2	Optimization Tips	12
11	Common Mistakes	12
11.1	Mistake 1: Confusing word-break and overflow-wrap	12
11.2	Mistake 2: Forgetting Language Tag	13
11.3	Mistake 3: Breaking Code Unintentionally	13
11.4	Mistake 4: Not Testing on Mobile	13
12	Advanced Techniques	14
12.1	Soft Hyphens	14
12.2	Custom Breaking Points	14
13	Real-World Comprehensive Example	14
14	Summary: Key Takeaways	16
14.1	Property Purposes	16
14.2	Best Practices	16
14.3	Quick Reference	16
15	Conclusion	17

1 Introduction

The CSS `word-wrap` and `word-break` properties control how text breaks when it reaches container boundaries. These properties are essential for handling long words, URLs, code snippets, and responsive layouts.

1.1 The Word Breaking Problem

Without proper word breaking controls:

- Long URLs overflow containers
- Email addresses break incorrectly
- Technical terms display unpredictably
- Mobile layouts become unusable
- Code snippets become unreadable

1.2 When Word Breaking Matters

- Mobile responsive design
- Chat applications with URLs
- Code blocks and terminals
- Email addresses and usernames
- Long technical documentation
- Multilingual content
- User-generated content

2 overflow-wrap Property

The `overflow-wrap` property (formerly `word-wrap`) controls whether to break words that exceed container boundaries.

2.1 overflow-wrap Values

Value	Behavior	Use Case
<code>normal</code>	Don't break words	Default, allow overflow
<code>break-word</code>	Break at any point if needed	Prevent overflow
<code>anywhere</code>	Break at any point	More aggressive breaking

2.2 overflow-wrap Examples

```
/* Default: allow overflow */
p {
  overflow-wrap: normal;
}

/* Break long words */
.chat-message {
  overflow-wrap: break-word;
  width: 300px;
}

/* Most aggressive breaking */
.strict-container {
  overflow-wrap: anywhere;
  width: 200px;
}

/* Common usage */
p {
  overflow-wrap: break-word;
  word-break: break-word; /* Older property */
}
```

2.3 word-wrap (Deprecated)

The `word-wrap` property is the older name for `overflow-wrap`:

```
/* Deprecated but still works */
p {
  word-wrap: break-word;
}

/* Modern equivalent */
p {
  overflow-wrap: break-word;
}

/* Use modern property for new projects */
```

3 word-break Property

The `word-break` property specifies breaking rules for words that don't fit.

3.1 word-break Values

Value	Behavior	Use
<code>normal</code>	Standard breaking rules	Default
<code>break-all</code>	Break at any character	Force fit
<code>keep-all</code>	Keep words together	CJK text
<code>break-word</code>	Legacy alias	Avoid, use <code>overflow-wrap</code>

3.2 word-break Examples

```
/* Standard breaking (hyphenation points) */
p {
  word-break: normal;
}

/* Break anywhere to fit */
.strict {
  word-break: break-all;
}

/* Keep words together (CJK) */
.cjk-text {
  word-break: keep-all;
}

/* Don't break words, allow overflow */
code {
  word-break: normal;
  overflow-x: auto;
}
```

4 Comparing word-break and overflow-wrap

Understanding the differences is crucial for proper implementation.

4.1 Behavior Differences

Scenario	<code>word-break</code>	<code>overflow-wrap</code>
Long word at line start	Breaks word	Moves to next line
Space available	Different	Prioritizes breaks at spaces
Hyphenation	Not affected	May apply hyphens
Mobile wrapping	More aggressive	More conservative

4.2 Detailed Examples

```
/* Text: "The URL is verylongwordthatdoesntfit" */
/* Container width: 200px */

/* word-break: normal (default) */
.normal {
  word-break: normal;
  /* verylongwordthatdoesntfit overflows */
}

/* word-break: break-all */
.break-all {
  word-break: break-all;
  /* verylongwordthatdoesntf
    it breaks at any point */
}

/* overflow-wrap: break-word */
.wrap {
  overflow-wrap: break-word;
  /* The URL is
    verylongwordthatdoesntfit
    breaks at space first */
}

/* overflow-wrap: anywhere */
.anywhere {
  overflow-wrap: anywhere;
  /* Similar to break-all
    but considers word breaks */
}
```

5 Real-World Use Cases

5.1 Use Case 1: Chat Application

```
.chat-message {
  max-width: 400px;
  padding: 10px;
  background-color: #e3f2fd;
  border-radius: 8px;
  word-wrap: break-word;
  overflow-wrap: break-word;
}

/* Long URL in chat */
.chat-message a {
  color: #0066cc;
  word-break: break-all; /* Break URLs aggressively */
}

.chat-message p {
  word-break: normal; /* Normal for regular text */
}
```

5.2 Use Case 2: Code Block Display

```
pre {
  background-color: #f4f4f4;
  padding: 15px;
  border-radius: 4px;
  overflow-x: auto; /* Horizontal scroll for long lines */
  word-break: normal; /* Don't break code */
  word-wrap: normal; /* Don't wrap code */
}

code {
  font-family: 'Courier New', monospace;
  word-break: normal;
  white-space: pre; /* Preserve formatting */
}

/* Inline code */
span.code {
  background-color: #f0f0f0;
  padding: 2px 4px;
  overflow-wrap: break-word; /* Break if too long */
}
```

5.3 Use Case 3: Email Address Display

```
.email {
  word-break: break-all; /* Allow breaking long emails */
}
```

```
    overflow-wrap: break-word; /* Prefer space breaks */
}

/* Better: use proper HTML and CSS */
<a href="mailto:verylongemailaddress@example.com">
    verylongemailaddress@example.com
</a>

<style>
a[href^="mailto:"] {
    word-break: break-all;
}
</style>
```

5.4 Use Case 4: Username Display

```
.username {
    word-break: break-all;
    max-width: 150px;
}

.username-long {
    word-break: break-all;
    font-weight: 600;
    color: #333;
}

/* Truncate alternative */
.username-truncated {
    max-width: 150px;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
```

5.5 Use Case 5: Responsive Container

```
.container {
    max-width: 100%;
    padding: 15px;
}

.content {
    font-size: 16px;
    line-height: 1.6;
    overflow-wrap: break-word; /* Fallback */
    word-break: break-word; /* Modern support */
}

/* Mobile breakpoint */
@media (max-width: 480px) {
```

```

.content {
  font-size: 14px;
  word-break: break-all; /* More aggressive on mobile */
}

/* Long URL handling */
a {
  word-break: break-all;
  overflow-wrap: break-word;
}

```

6 Combining with white-space

The `white-space` property affects word breaking behavior.

6.1 white-space Interactions

white-space	Word Breaking	Result
normal	Depends on break property	Standard behavior
nowrap	Ignored	No wrapping occurs
pre	Ignored	Breaks only at newlines
pre-wrap	Applied	Wraps + preserves spaces

6.2 Combining Examples

```

/* Allow wrapping with space preservation */
pre {
  white-space: pre-wrap;
  word-break: break-word;
  overflow-wrap: break-word;
}

/* Preserve newlines, break long lines */
code {
  white-space: pre-wrap;
  word-break: break-all;
}

/* Single line, no wrapping */
.single-line {
  white-space: nowrap;
  word-break: normal; /* Ignored due to nowrap */
  overflow: hidden;
  text-overflow: ellipsis;
}

/* Formatted text with wrapping */
.formatted {
  white-space: pre-wrap;
  overflow-wrap: break-word;
}

```

```
}
```

7 Hyphens and word-break

The `hyphens` property works alongside word breaking.

7.1 Hyphens Integration

```
/* Automatic hyphenation */
p {
  hyphens: auto;
  overflow-wrap: break-word;
  lang: en; /* Required for hyphenation */
}

/* Manual hyphenation points */
p {
  hyphens: manual;
  overflow-wrap: break-word;
}

/* No hyphenation */
p {
  hyphens: none;
  word-break: break-word;
}

/* Combined approach */
article {
  hyphens: auto;
  overflow-wrap: break-word;
  text-align: justify;
  lang: en;
}
```

8 CJK (Chinese, Japanese, Korean) Text

Special considerations for CJK languages.

8.1 CJK-Specific Properties

```
/* Don't break CJK words */
.cjk-text {
  word-break: keep-all;
}

/* Allow breaking anywhere for space saving */
.cjk-strict {
  word-break: break-all; /* More aggressive */
}
```

```
}

/* Balanced approach */
.cjk-balanced {
  word-break: normal; /* Standard CJK breaking */
  overflow-wrap: break-word;
}

/* Japanese text specific */
.japanese {
  word-break: keep-all; /* Keep words together */
  lang: ja;
}

/* Chinese text specific */
.chinese {
  word-break: break-all; /* Can break anywhere */
  lang: zh;
}
```

9 Browser Support

9.1 overflow-wrap and word-break Support

Property	Chrome	Firefox	Safari	Edge
overflow-wrap	23+	49+	6.1+	All
word-wrap	All	All	All	All
word-break	All	15+	All	All
word-break: break-word	All	15+	All	All

9.2 Fallback Strategies

```

/* Modern with fallback */
p {
  /* Fallback for older browsers */
  word-wrap: break-word;

  /* Modern property */
  overflow-wrap: break-word;
}

/* Feature detection */
@supports (overflow-wrap: break-word) {
  p {
    overflow-wrap: break-word;
  }
}

@supports not (overflow-wrap: break-word) {
  p {
    word-wrap: break-word;
  }
}

/* Comprehensive approach */
.container {
  word-wrap: break-word; /* IE 5.5+ */
  overflow-wrap: break-word; /* Modern browsers */
  word-break: break-word; /* Alternative */
}

```

10 Performance Considerations

10.1 Performance Impact

- **word-break: normal** - No performance cost
- **word-break: break-all** - Minimal impact
- **overflow-wrap: break-word** - Minimal impact
- **hyphens: auto** - Can impact performance (requires language detection)

- **Large text blocks** - May need optimization

10.2 Optimization Tips

```
/* Only apply breaking where needed */
.user-content {
  overflow-wrap: break-word;
}

/* Not on everything */
body {
  /* Don't set globally */
}

/* Combine with truncation */
.long-text {
  max-width: 400px;
  overflow: hidden;
  text-overflow: ellipsis;
  white-space: nowrap;
  /* Breaking not needed with truncation */
}

/* Cache language setting */
html {
  lang: en; /* Set once at root */
}

article {
  hyphens: auto; /* Works with root lang */
}
```

11 Common Mistakes

11.1 Mistake 1: Confusing word-break and overflow-wrap

Bad:

```
p {
  overflow-wrap: break-all; /* Too aggressive */
}
```

Good:

```
p {
  overflow-wrap: break-word; /* Better default */
}

/* For aggressive breaking */
code {
  word-break: break-all; /* Explicit intent */
}
```

11.2 Mistake 2: Forgetting Language Tag

Bad:

```
p {  
  hyphens: auto;  
  /* No lang attribute - hyphens don't work */  
}
```

Good:

```
<!-- HTML -->  
<html lang="en">  
  <p>Text here...</p>  
</html>  
  
<!-- CSS -->  
p {  
  hyphens: auto; /* Works with lang attribute */  
}
```

11.3 Mistake 3: Breaking Code Unintentionally

Bad:

```
pre {  
  word-break: break-all;  
  /* Code breaks mid-word */  
}
```

Good:

```
pre {  
  white-space: pre;  
  word-break: normal;  
  overflow-x: auto; /* Horizontal scroll instead */  
}
```

11.4 Mistake 4: Not Testing on Mobile

Bad:

```
p {  
  overflow-wrap: normal; /* May overflow on mobile */  
}
```

Good:

```
p {  
  overflow-wrap: break-word; /* Safe for all sizes */  
}  
  
@media (max-width: 480px) {  
  p {  
    word-break: break-all; /* More aggressive if needed */  
  }  
}
```

```
}  
}
```

12 Advanced Techniques

12.1 Soft Hyphens

Use soft hyphens to control breaking points:

```
<!-- HTML with soft hyphens -->  
<p>superfluous&shy;word&shy;with&shy;breaks</p>  
  
<style>  
p {  
  hyphens: manual; /* Respects &shy; -->  
}  
</style>  
  
<!-- Only breaks at soft hyphen if needed -->
```

12.2 Custom Breaking Points

```
/* Using word-spacing for custom breaks */  
.custom-breaks {  
  word-spacing: 0;  
  overflow-wrap: break-word;  
}  
  
/* Breaking long URLs */  
a {  
  word-break: break-all;  
  overflow-wrap: break-word;  
}  
  
/* Email addresses */  
.email {  
  word-break: break-all;  
}  
  
/* Path-like content */  
.path {  
  word-break: break-all;  
  font-family: monospace;  
}
```

13 Real-World Comprehensive Example

```
/* Chat application styling */  
.chat-container {  
  max-width: 600px;
```

```
    margin: 0 auto;
}

.message {
    padding: 12px 16px;
    margin-bottom: 8px;
    border-radius: 8px;
    word-wrap: break-word;
    overflow-wrap: break-word;
}

.message p {
    margin: 0;
    line-height: 1.4;
}

/* User message */
.message.user {
    background-color: #007bff;
    color: white;
    border-radius: 18px 4px 18px 18px;
}

/* System message */
.message.system {
    background-color: #e9ecef;
    color: #333;
}

/* Links in messages */
.message a {
    color: #0066cc;
    word-break: break-all; /* Break long URLs */
    text-decoration: underline;
}

.message.user a {
    color: #b3d9ff;
}

/* Code in messages */
.message code {
    background-color: rgba(0, 0, 0, 0.1);
    padding: 2px 4px;
    border-radius: 3px;
    word-break: break-all;
    font-family: monospace;
    font-size: 0.9em;
}

/* Username display */
.message .username {
    font-weight: 600;
    font-size: 0.85em;
}
```

```
word-break: break-all;
margin-bottom: 4px;
}
```

14 Summary: Key Takeaways

14.1 Property Purposes

Property	Purpose	Use
overflow-wrap	Break words at container edge	Default choice
word-break	Set breaking rules	Special cases
white-space	Preserve formatting	Code blocks
hyphens	Add hyphenation	Long text

14.2 Best Practices

1. Use `overflow-wrap: break-word` as default
2. Use `word-break: break-all` for aggressive breaking
3. Don't break code; use horizontal scroll
4. Set language tag for hyphenation
5. Test on mobile devices
6. Combine properties appropriately
7. Use soft hyphens for control
8. Consider CJK text needs
9. Prefer `overflow-wrap` over deprecated `word-wrap`
10. Provide fallbacks for older browsers

14.3 Quick Reference

```
/* Default safe choice */
p {
  overflow-wrap: break-word;
}

/* For chat/messages */
.message {
  overflow-wrap: break-word;
  word-break: break-word;
}

/* For URLs */
a {
  word-break: break-all;
}
```

```
}

/* For code */
pre {
  white-space: pre-wrap;
  word-break: normal;
  overflow-x: auto;
}

/* For justified text */
p {
  text-align: justify;
  hyphens: auto;
  overflow-wrap: break-word;
}

/* Mobile responsive */
@media (max-width: 480px) {
  p {
    word-break: break-word;
  }
}
```

15 Conclusion

CSS word-wrap and word-break properties are essential tools for creating responsive, readable web content. Understanding when and how to apply these properties ensures your text displays correctly across all devices and contexts.

Key principles to remember:

1. **overflow-wrap** controls word wrapping at container edge
2. **word-break** sets breaking rules for words
3. Always test on mobile
4. Preserve code formatting with horizontal scroll
5. Set language tag for hyphenation
6. Combine properties for optimal results
7. Use soft hyphens for control
8. Provide fallbacks for older browsers
9. Consider language-specific needs
10. Balance readability with aesthetics

With mastery of word-wrap and word-break properties, you'll create responsive layouts that handle all types of content elegantly, improving user experience across all devices.

Happy coding!